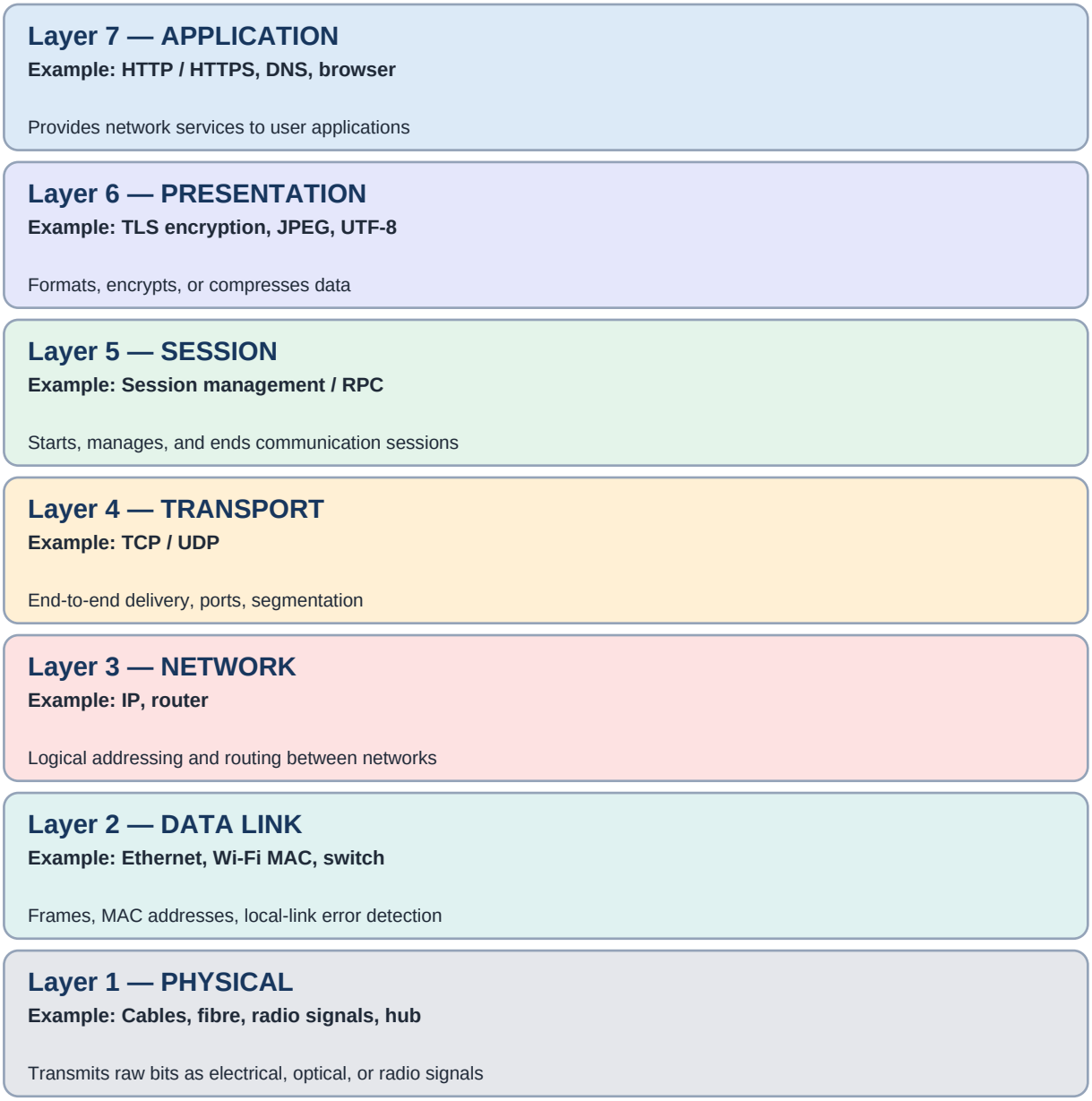
OSI 7-LAYER MODEL

Question 1 — Diagram and examples of all seven layers

The OSI (Open Systems Interconnection) model is a framework that explains how data travels from one device to another over a network. It divides communication into seven layers. Each layer has a different job and works with the layers above and below it.



Data is encapsulated as it moves down the sender's stack and decapsulated at the receiver.

Remember the order: Application, Presentation, Session, Transport, Network, Data Link, Physical. When sending data, it moves from Layer 7 down to Layer 1; at the receiving device, it is processed from Layer 1 up to Layer 7.

Question 2 — Match each scenario to the OSI layer

Scenario	Correct answer	Reason
a) Encrypting data before sending	**Presentation Layer (Layer 6)**	This layer represents data formatting, encryption, and compression. In real networks, TLS encryption is commonly associated with the presentation function in the OSI model, although practical TCP/IP stacks do not always map neatly to one layer.
b) Routing packets across networks	**Network Layer (Layer 3)**	Routers use IP addresses and routing tables to forward packets between different networks.
c) Converting data to electrical signals	**Physical Layer (Layer 1)**	The physical layer transmits raw bits using electrical, optical, or radio signals through copper cables, fibre, or wireless links.
d) Opening a web page in your browser	**Application Layer (Layer 7)**	The browser uses application protocols such as HTTP or HTTPS to request and receive web content. Other layers still carry the data underneath.
e) Detecting and correcting errors in frames	**Data Link Layer (Layer 2)**	The data link layer handles frames and commonly detects local-link errors using a frame check sequence (FCS). Some link technologies also support retransmission or correction; error correction can also occur at other layers.

Quick answer list in the requested format

• a) Encrypting data before sending: Presentation Layer (Layer 6)

• b) Routing packets across networks: Network Layer (Layer 3)

• c) Converting data to electrical signals: Physical Layer (Layer 1)

• d) Opening a web page in your browser: Application Layer (Layer 7)

• e) Detecting errors in frames: Data Link Layer (Layer 2)

Question 3 — How OSI layers work when using WhatsApp

Example: sending a text message on WhatsApp. WhatsApp uses the internet and several protocols together. The OSI model is a learning framework, so real protocols do not always fit perfectly into only one layer.

Application Layer (Layer 7)

The WhatsApp app provides the user-facing function of sending a message, choosing a contact, and showing delivered or read status. It creates the message request and communicates with WhatsApp's service using application-level protocols.

Presentation Layer (Layer 6)

Message data must be represented in a format that the sender and receiver can process. Encryption protects message contents in transit. WhatsApp uses end-to-end encryption for personal messages; encryption is often taught under the presentation layer, although actual implementations may place cryptographic functions across application and transport-related components.

Transport Layer (Layer 4)

Transport protocols carry data between applications and help manage delivery. TCP provides ordered, reliable delivery for many internet services; UDP is used by other real-time services. The exact transport details used by an app can vary, but this layer is responsible for transport between endpoints and uses port numbers.

Network Layer (Layer 3)

IP addresses help route packets from the phone across the mobile network or Wi-Fi network and through internet routers toward the service. Routers examine network-layer information to choose the next path.

Data Link and Physical Layers (Layers 2 and 1)

On Wi-Fi, the data link layer moves frames between the phone and the access point using local-link addressing. The physical layer sends bits as radio signals. On a mobile connection, the radio access network performs equivalent lower-layer functions using cellular technology.

What happens step by step

• You type a message and tap Send in WhatsApp (application function).

• The message is represented and protected by encryption before it is transmitted.

• Transport functions carry the data between the relevant endpoints.

• IP packets are routed through the Wi-Fi/mobile network and the internet.

• Local network frames and radio signals carry the bits from the phone toward the network.

• The recipient's device processes the incoming data up through the layers and WhatsApp displays the message.

Question 4 — Website fails, but ping to its IP address works

Most likely layers to investigate: Application (Layer 7), Transport (Layer 4), and possibly DNS-related application/name-resolution configuration. The Network layer is at least partly working because the IP address responds to ping, but a successful ping does not prove that the website service is working.

Reasoning

Ping usually sends ICMP messages to test IP reachability. A website normally uses HTTP or HTTPS, commonly over TCP port 80 or 443 (or another supported transport arrangement). Therefore, a host may answer ping while its web server is down, its web port is blocked, or the browser cannot resolve the domain name correctly.

Step-by-step troubleshooting

Step 1: Check the exact URL. Confirm the spelling and try the website by domain name and, if appropriate, a known correct URL. A wrong URL can cause a browser-level failure.

Step 2: Check DNS/name resolution. If the site works by IP but not by domain name, DNS may be failing or returning an incorrect address. Note that many websites use virtual hosting and HTTPS, so entering the IP alone may not load the correct site.

Step 3: Test HTTP/HTTPS access. Try another browser or device and check whether the browser reports a timeout, certificate error, or name-resolution error. A timeout may indicate a blocked port or unavailable web service.

Step 4: Check firewall and proxy settings. A local firewall, company proxy, VPN, or network filter may block web traffic while still allowing ICMP ping.

Step 5: Check the web server/service. If other users also cannot open the site, the web server or hosting service may be down, misconfigured, or not listening on the expected port.

Step 6: Compare with another website. If other sites open normally, the issue may be specific to that domain, its DNS record, certificate, or server. If no websites open, check the device's broader network and proxy configuration.

Conclusion: A successful ping confirms that the target IP responds to ICMP, but it does not confirm that DNS, HTTP/HTTPS, the browser, or the website server is working. Start with DNS and application/web-service checks, then investigate transport ports and filtering. The exact cause depends on the error message.